Network Engineering Portfolio: Purpose, Goals and Roadmap

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Why I am sharing these labs

I am building hands-on skills as a network engineer and designer. The Packet Tracer labs in this portfolio are my working record of that development: what I learned, how I applied it, what broke, and how I fixed it.

I am sharing them for three reasons:

- **To show progress honestly.** The early labs are proofs of concept and are presented as such. The larger builds show the same ideas redesigned with more care, validation and security.
- **To show how I think, not only what I built.** Each overview records the design decisions, the testing and the troubleshooting, including mistakes I found and corrected.
- **To get feedback.** I want instructors, peers and reviewers to point out gaps so I can improve the next build.

How the labs fit together

Group	Files	Intent
Proof-of-concept labs	Five practice labs: OSPF-EIGRP / EIGRP-BGP campus, Intranetwork V2.0, Network Data Infrastructure LAN, Project Network, Completed final	After learning a technology, build a small network to apply it, see how it behaves and practice designing around it
Three-Tier Campus	Campus_Network	Apply the three-tier model (access, distribution, core) with HSRPv2, multilayer switching, OSPF, EIGRP and BGP, GRE and a services segment

Group	Files	Intent
CollapseCore Campus	CollapseCore_CampusNetwork	A collapsed-core design taken further: validate and harden it with HSRP tracking, STP root placement, RADIUS AAA and a zone-based firewall

The proof-of-concept labs are deliberately narrow, with one or two concepts each. The two campus builds combine those concepts into complete designs.

The Three-Tier build is documented as it stands, including the gaps found by reading its configuration. CollapseCore shows how I closed comparable gaps through testing and troubleshooting.

How I work and improve

Every lab follows the same cycle, and each pass through it raises the standard for the next build.

1. **Learn** the technology: what problem it solves and how it works.
2. **Build** a small lab to apply it.
3. **Verify** with real tests: ping, traceroute, `show` commands and failover tests, not only a clean configuration.
4. **Troubleshoot** failures and record the cause and fix.
5. **Harden** with SSH only, AAA, port security, spanning-tree protection and firewalling.
6. **Document** the design, the addressing, the tests and the open items.

Habits I am building

- Validate every address, tunnel endpoint and gateway before calling a build done.
- Use unique addresses and a written addressing plan.
- Treat a feature as finished only when it has been tested, including failure cases.
- Keep a troubleshooting log so the same mistake is not repeated.

Building troubleshooting skills

These labs are also a deliberate way to get faster and more accurate at troubleshooting. Real faults, such as a DHCP failure, a tunnel that never came up or traffic that was slow, are better teachers than a lab that works the first time, so I keep a troubleshooting log for every build and treat each fault as practice.

My method

1. **Define the symptom** precisely: what fails, from where, and what still works.
2. **Work up the layers:** check physical and link state, then VLANs and trunks, then addressing and gateways, then routing, then services and security policy.
3. **Isolate the scope** with ping and traceroute from different sources, such as the host, the gateway and the next hop.
4. **Use the right `show` command** for the layer, such as `show ip interface brief`, `show vlan brief`, `show interfaces trunk`, `show ip route`, `show ip ospf neighbor`, `show standby brief` and `show ip dhcp binding`.
5. **Change one thing at a time**, then retest.
6. **Record** the symptom, root cause and fix.

How I will measure improvement

- Time to identify the root cause, tracked per fault in the log.
- Whether the first hypothesis was the right one, so I learn to check the most likely cause first.
- Fewer repeat faults, such as mistyped addresses, as validation becomes habit.
- Practicing without hints: breaking a working lab on purpose, or having someone else break it, and fixing it from the CLI alone.

Goals

Certification: work toward CCNP Enterprise, which requires the core exam (ENCOR) and one concentration exam. The labs already exercise much of the core content: dual-stack addressing, OSPF, EIGRP and BGP, switching and spanning tree, first-hop redundancy, GRE tunnels and device security.

Higher education: continue my studies at Rowan University and keep tying coursework to hands-on builds, so theory and practice reinforce each other.

As an engineer: be able to configure, verify and troubleshoot an enterprise network quickly and accurately, from a CLI alone.

As a designer: produce designs that are documented, scalable and secure from the start, with sound reasoning behind each choice such as why a three-tier or collapsed-core model, where the failure domains are, and how traffic is meant to flow.

Over time: move from labs that prove one concept to complete designs that could stand up to a real review.

Staying current: protocols, standards and good practice

Fundamentals stay at the center, and newer technology is added on top of them.

Area	Fundamentals I keep sharp	Newer or next-step topics
Layer 2	IEEE 802.1Q VLANs, spanning tree, 802.1AX/LACP EtherChannel	Rapid PVST+ and MST, 802.1X port-based access control
Routing	OSPF, EIGRP, eBGP, redistribution	IPv6 with OSPFv3, EIGRP named mode, BGP policy
First-hop redundancy	HSRP with tracking	HSRP for IPv6, VRRP
Security	SSH, AAA with RADIUS/TACACS+, zone-based firewall	Device hardening baselines, segmentation, WPA3 for wireless
Wireless	WLC and AP basics	Controller-based design and IEEE 802.11 standards
Operations	Syslog, SNMP, NTP	Network automation (Python, NETCONF/RESTCONF), configuration templates
Design	Three-tier and collapsed-core models	Cisco validated design guidance, SD-Access and SD-WAN concepts

Good practice I follow: follow Cisco design guidance and the relevant RFCs and IEEE standards; use a separate management VLAN; keep unused ports and services off; keep the console local so a AAA outage cannot lock me out; and document every change.

Roadmap

Next for the labs

- Fix the findings in the Three-Tier build (the two GRE destination typos, duplicate SVI addresses and DHCP gateways) and apply the CollapseCore hardening.
- Run and record failover and reachability tests for both campus builds.
- Close the open CollapseCore items: NTP through the firewall, DHCP pool ranges and the Guest limit, and the CME cleanup.
- Revisit the early labs to correct their small addressing issues, such as shared tunnel addresses, as short refresh exercises.
- Add a troubleshooting practice set: deliberately broken versions of my labs, each with a time-to-fix target.

Next for the learning path

- Add IPv6 and OSPFv3 to a new lab.
- Practice Rapid PVST+ or MST and 802.1X.
- Start a small automation lab (Python against device configurations).
- Map completed lab work to the CCNP Enterprise exam topics and schedule study around the gaps.

Next for sharing

- Keep one overview per project in the same format and link them from a single portfolio page, ordered from early practice to the hardened build.